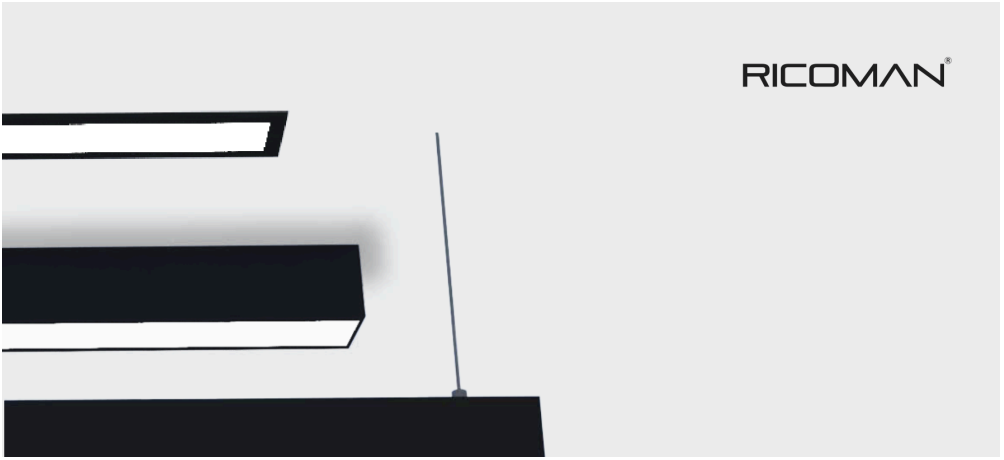


Flow+

User Manual For Surface & Suspended Installation

Please read these instructions carefully before installation and leave a copy with the end user for future reference.

Watch the installation video

Scan the QR code with your phone camera, or go to

ricoman.com/flow-install

A quick 2-minute guide, plus the full step-by-step.

WARNING: never connect the LED strip straight to mains (230V). It must be powered through the LED driver supplied. Mains goes to the driver only. Connecting mains to the strip destroys it at once and voids the warranty. **Switch off and isolate the power before you start.** Keep it off until the LED strip is fully installed and connected, and you reach the test step.

Start here

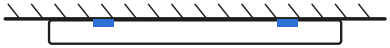
Watch the installation video

Scan the QR code with your phone camera, or go to

ricoman.com/flow-install

A quick 2-minute guide, plus the full step-by-step.

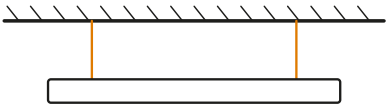
How are you mounting it?



Surface

Fixed straight to the ceiling with clips

Section 3A · page 7



Suspended

Hanging on wires

Section 3B · page 9

Contents

1	Before you start	4
2	Build the profile shape	6
3A	Surface: fix to the ceiling	7
3B	Suspended: hang it on wires	9
4	Roll out the LED strip	15
5	Connect the power	17
6	Test and fit the diffuser	20
7	Seal the joints	22
8	Final check and troubleshooting	23

Everyone follows sections 1 and 2, then **either 3A or 3B**, then 4 to 8.

Safety and important information

SAFETY INFORMATION

Install by a fully qualified and competent electrician who understands building regulations and BS 7671 (IET Wiring Regulations). Before commencing any installation or maintenance work, switch off and lock off the mains.

Flow+ runs on 24V from a separate driver. Never connect mains (230V) to the fitting or the input lead.

IMPORTANT USER INFORMATION

Install strictly in accordance with these instructions, with all the tools needed. Improper installation could cause harm to property or persons and voids your 5-year warranty. It is the installer's responsibility to consider fire and safety risk and take appropriate precautions.

THE FIVE RULES

1. Never connect mains to the fitting. 2. Build and mount the profile before the LED strip goes in. 3. Test before you seal the joints. 4. Never stretch the diffuser. 5. Never let the flex carry weight.



PRE INSTALLATION INSTRUCTIONS & GUIDELINES

• Cross-reference every part with the delivery note within 3 working days. Report shortages or damage with photos.

• Do not introduce any parts other than those provided, or cut, drill or modify the profile, tape or diffuser except as shown here.

• Fix only to a structure that can hold more than five times the weight of the fitting.

• IP20: indoor, dry locations, 0°C to 45°C. Not above hot objects such as ovens.

• LED chips can be damaged by gases and vapours from materials such as rubber.

• Leave space for ventilation around the driver. Do not cover with insulation.

• Ensure all electrical connections are tight with no loose strands.

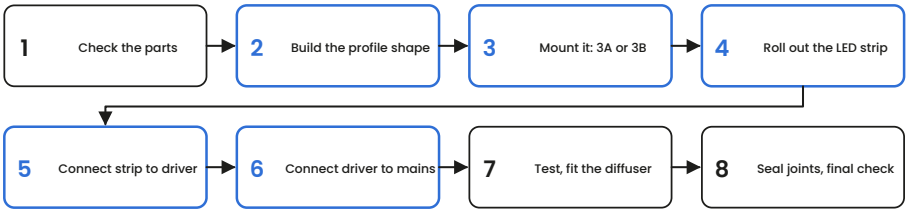
• Not suitable for highly flammable surfaces (e.g. polystyrene, textiles).

• Before making fixing holes, check for hidden pipes and cables.

• Only power the LED strip through the LED driver supplied for this fitting.

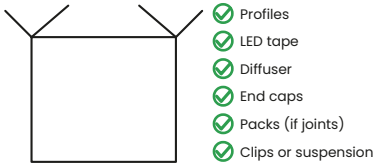
1 Before you start

WARNING: never connect the LED strip straight to mains (230V). It must be powered through the LED driver supplied. Mains goes to the driver only. Connecting mains to the strip destroys it at once and voids the warranty.
Switch off and isolate the power before you start. Keep it off until the LED strip is fully installed and connected, and you reach the test step.



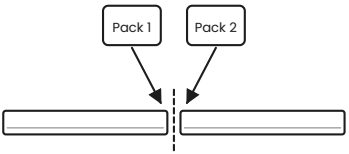
The order of work. Build the whole profile shape and mount it first. Only then roll the LED strip out, connect it to the driver, and connect the driver to mains.

1



Cross-reference every part with the delivery note within 3 working days: profiles, LED tape, diffuser, end caps, packs and fixings. Report anything missing with photos.

2

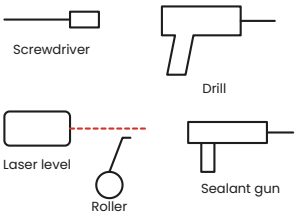


2 packs at every joint

Runs with joints: **2 connection packs per joint** (R383535-U-03 for W35). A single section has no joints and needs none.

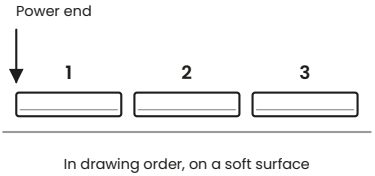
1 Before you start (continued)

3



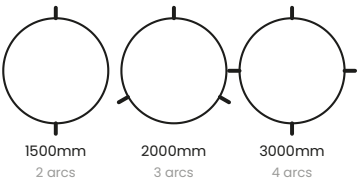
Tools: screwdriver, Allen keys, drill, laser level, sealant gun, clean dry cloth, hand roller, cable ties, multimeter.

4



Lay every section out on a clean, soft surface **in drawing order**, starting from the power end. Check each label.

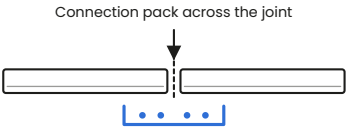
5



Rings up to 1200mm come complete with a ceiling rose. Larger rings come as arcs for you to join.

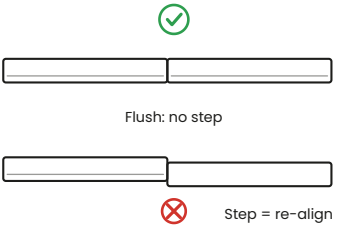
2 Build the profile shape

1



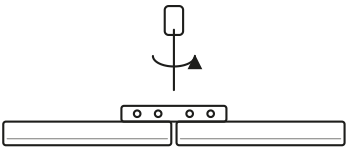
Fit the **connection pack** across the joint: **2 packs per joint**. Put the screws in, **not tight yet**.

2



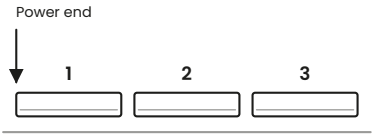
Run your finger over the top and sides. There must be **no step**.

3



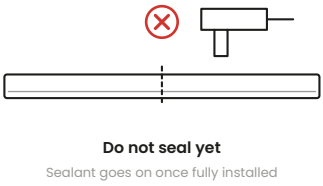
Tighten every connection pack screw.

4



Repeat for every joint until the whole shape is one piece. It is mounted next; the LED strip goes in after.

5

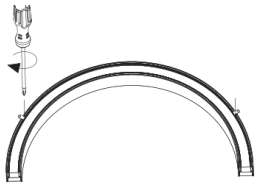


Do not seal the joints yet. Sealant goes on once the fitting is fully installed (section 7).

3A Surface: fix to the ceiling

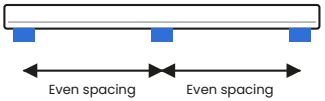
Surface mounting only. Hanging it on wires? Skip to **3B on page 9**. When this section is done, go to **section 4 on page 15**.

1



Fit the surface clips R38-03 to the back of the built profile.

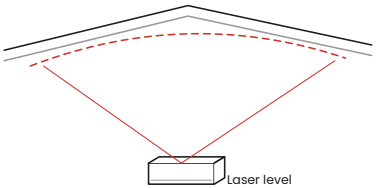
2



One clip close to each end

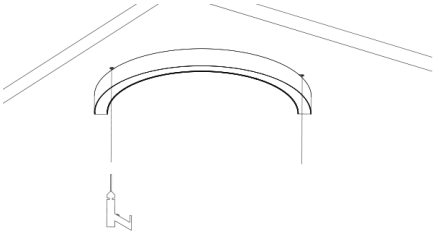
Space the clips evenly along the profile, with one close to each end.

3



Mark the run on the ceiling with a laser line.

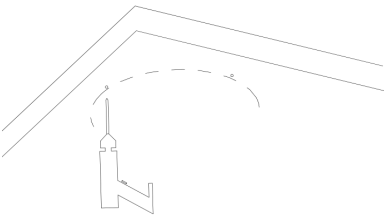
4



Hold the profile on the line and mark through each clip.

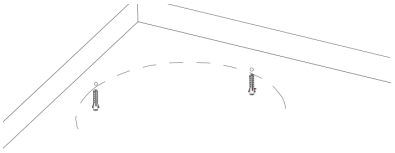
3A Surface: fix to the ceiling (continued)

5



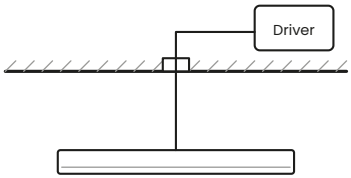
Take it down. Drill at each mark, checking for hidden pipes and cables.

6



Fit plugs to suit the ceiling: plasterboard, timber or concrete.

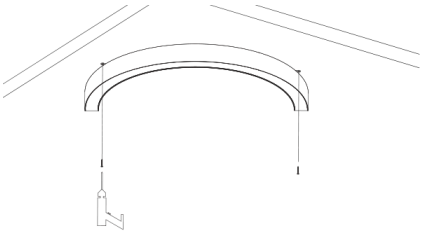
7



Feed the input lead through to the driver

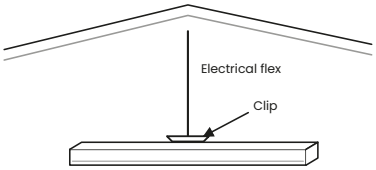
Feed the input lead through the ceiling to the driver position.

8



Line up the holes and screw the profile to the ceiling. Check it is straight against the laser line.

9



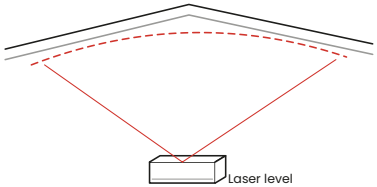
Clip the flex to the profile at the feed point

Clip the electrical flex to the profile at the feed point.

3B Suspended: prepare the ceiling

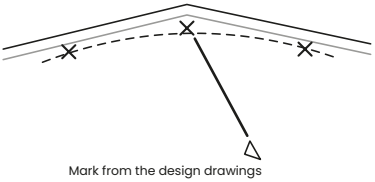
Suspended (on wires) only. Surface mounting? Use **3A on page 7**, then go to **section 4 on page 15**.

1



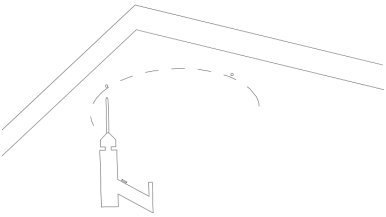
Mark the line of the run on the ceiling with a laser level.

2



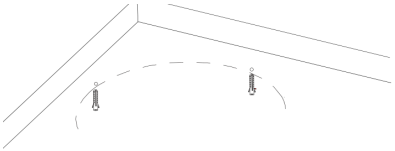
Mark each suspension point **as shown on your design drawings**.

3



Drill at each mark, checking for hidden pipes and cables.

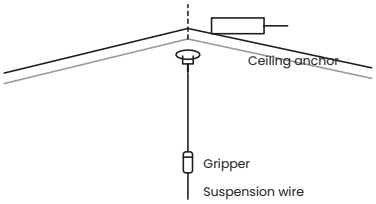
4



Fit plugs to suit the ceiling: plasterboard, timber or concrete.

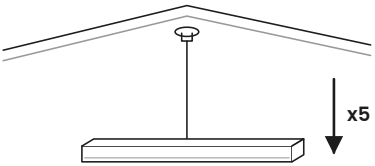
3B Suspended: prepare the ceiling (continued)

5



Fix a suspension anchor at each point (kit R3835-04 / R3850-04 / R3870-04 by profile width).

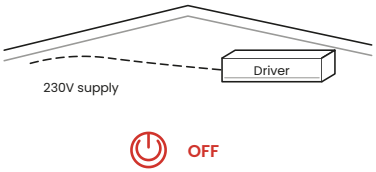
6



Each fixing holds 5x the weight

Check every fixing can hold 5 times the weight it carries.

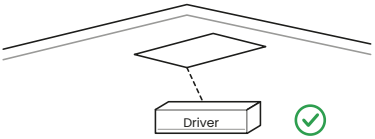
7



Isolated until the strip is in

With the power **isolated**, run the 230V supply to each driver position. Add a permanent live if there is an emergency pack. **Keep it switched off** until the LED strip is installed.

8

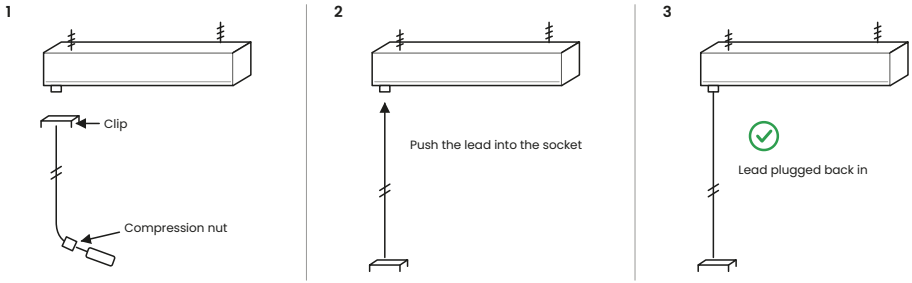


Reachable later: driver box or above an access panel

Put the driver where it can be reached later, such as above an access panel. **Using a Ricoman driver box (R38-05) or recessed ceiling plate (R38-06)?** See Suspended: fit the feed point, options A and B, in this section.

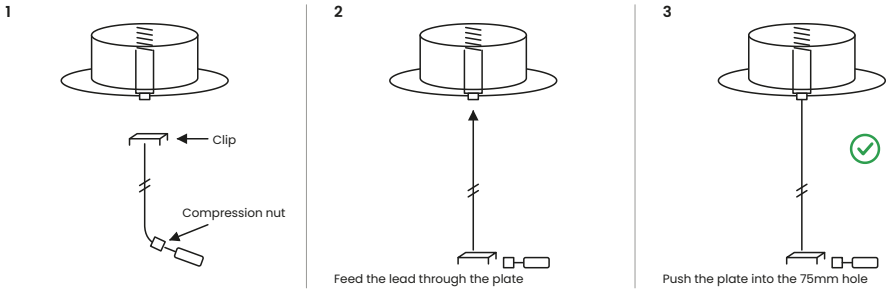
3B Suspended: fit the feed point (choose one)

A



Ceiling driver box R38-05: take the supply lead out, push the clip onto one end, plug the lead back into the box.

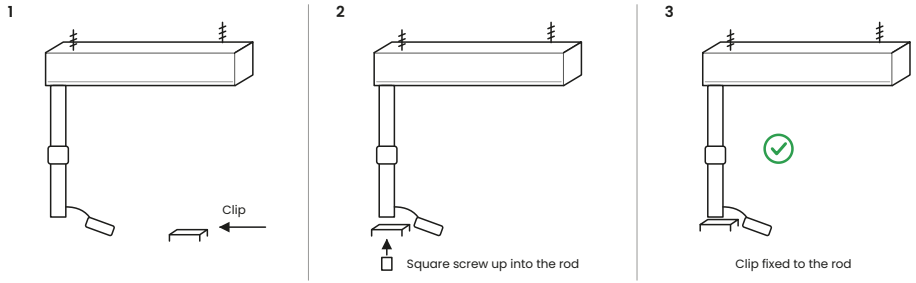
B



Recessed ceiling plate R38-06: cut a 75mm hole, clip the lead, feed it through the plate, push the plate into the hole.

3B Suspended: fit the feed point (choose one) (continued)

C



Rigid drop, mounting rod R38-07 (with A or B): put the square screw through the clip into the hole in the bottom of the rod.

3B Suspended: hang the profile

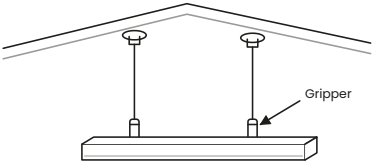
1



Enough people to spread the load

Lift the built profile with **enough people to spread the load evenly** along its length. Hold the profile, never the electrical flex.

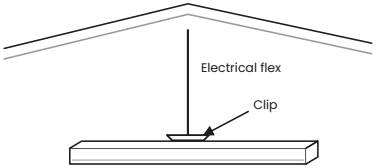
2



Clip onto every suspension wire

Clip the profile onto every suspension wire.

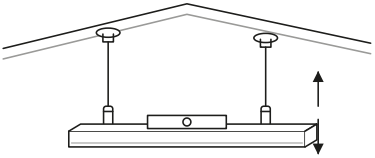
3



Clip the flex to the profile at the feed point

Clip the electrical flex to the profile at the feed point.

4

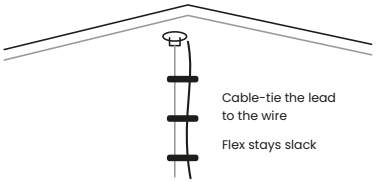


Pull the wire to raise, press the gripper to lower

Check level with the laser

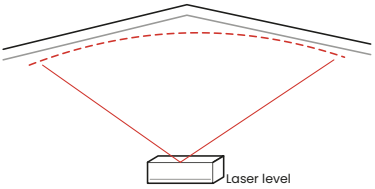
Level it: pull a wire through its gripper to raise; press the gripper top to lower.

5



Cable-tie the input lead to its suspension wire so it hangs slack.

6

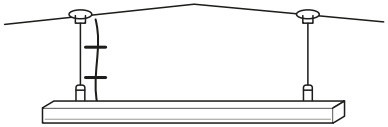


Check it is level and straight against the laser line before going on.

3B Suspended: hang the profile (continued)

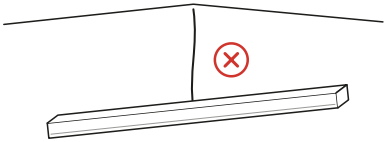
7

RIGHT



Wires carry the weight. Electrical flex tied on, slack

WRONG

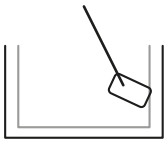


Never hang it by the electrical flex: the joint breaks

The suspension wires carry the weight. The electrical flex never does.

4 Roll out the LED strip

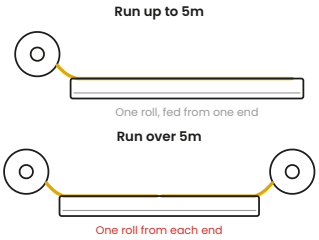
1



Channel dry and free of dust
Wipe with a clean dry cloth

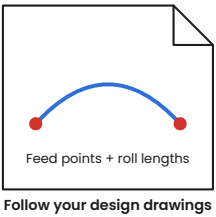
With the profile mounted, make sure the channel is **dry and free of dust**. Wipe it with a clean dry cloth if needed.

2



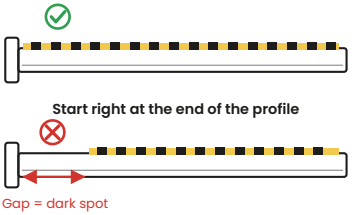
LED strip comes in rolls of **up to 5m**. Longer runs are supplied with more than one roll: **one roll from each end**.

3



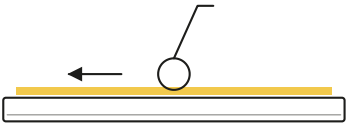
Check your design drawings for where each roll starts and where each end is powered.

4



Start the strip **as close to the end of the profile as possible**. A gap at the end shows as a dark spot.

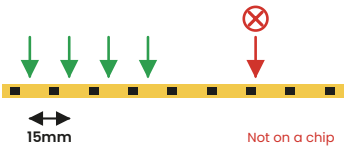
5



Peel a little backing, roll it out
From the start of the profile

Peel back a little of the backing and roll the strip out along the profile.

6



Press between the chips, never on them

Press the strip down **every 15mm** as you go. Press between the LED chips, **never on a chip**.

4 Roll out the LED strip (continued)

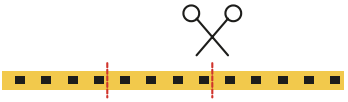
7



Let it follow the curve. Do not twist or stretch

The strip is **Z-bend tape**, made to arc round curves. Let it follow the curve; do not twist or stretch it.

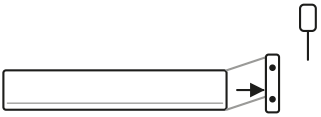
8



Cut only on a marked cut line

If you need to cut it, **cut only on a marked cut line**. Cut lines are marked on the strip.

9



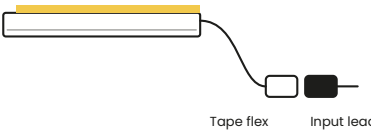
Push the end cap on and tighten its screws

Open ends only. A closed ring has none

Fit end caps on open ends and tighten the screws. A closed ring has no end caps. **Never pull on the flex**: its joint to the strip breaks easily.

5 Connect the power

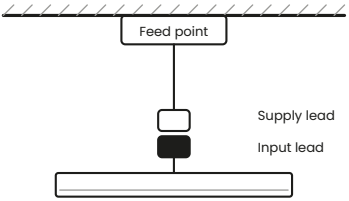
1



Plug the tape flex into the input lead (R38-08)

First, strip to driver: plug the strip flex into the input lead (R38-08) at each feed end.

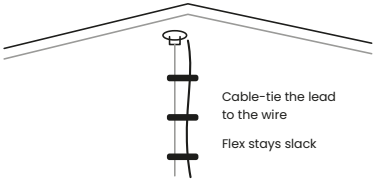
2



Plug the fitting input lead into the supply lead

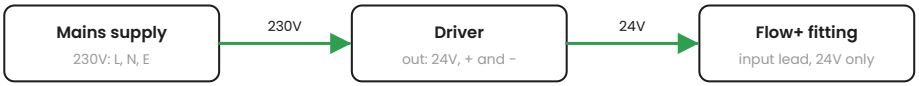
Connect the input lead to the driver 24V output, through the feed point.

3



Leave the flex slack. Never pull it tight or let it carry weight.

4



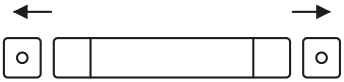
NEVER connect 230V straight to the fitting or the input lead

It destroys the LED strip at once and voids the warranty

Then, driver to mains (power still isolated): connect L, N and E to the driver input only. Use the driver supplied with the order. **Emergency pack ordered?** Wire it as shown on the next page.

5 Connect the power: emergency pack (only if ordered)

1

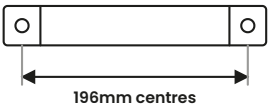


One screw in each end cap

Keep the caps safe to refit

With the power **isolated**, take off the two end caps on the pack.

2



Screws no bigger than 4mm

Fix the pack near the driver with two screws through the mounting points.

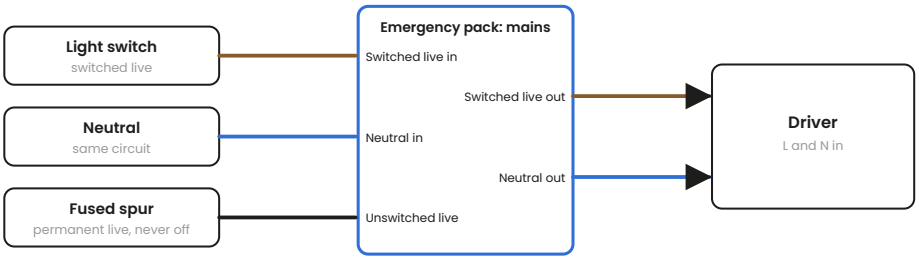
3



Check the label on the pack

Basic packs need a **key test switch**. Self-test and DALI packs test themselves and need none.

4

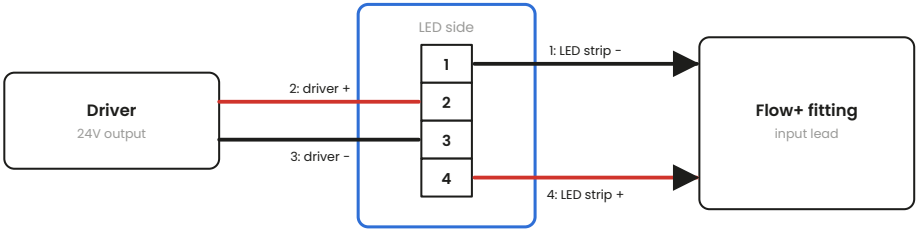


Use 1.0mm² cable minimum. The pack is double insulated: no earth needed.

Mains side. The light switch live and neutral go through the pack to the driver. Add a **permanent, unswitched live** from a fused spur on the same circuit as the lights.

5 Connect the power: emergency pack (only if ordered) (continued)

5

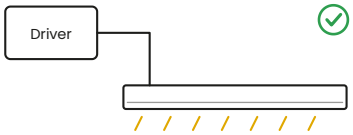


The driver feeds the pack. The pack feeds the fitting. Use 0.75mm² cable minimum.

LED side. Driver output into terminals 2 (+) and 3 (-). Fitting input lead from terminals 4 (+) and 1 (-). Then refit both end caps.

6 Test and fit the diffuser

1



Whole run lights evenly, no dim end

Fix any fault before sealing

Only now switch the power back on and test. The whole run must light evenly, with no flicker and no dim end. Fix any fault now (see the last page).

2

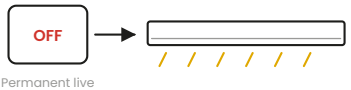


Steady green: basic pack charging

Flashing green: self-test or DALI pack

Emergency only: with the permanent live on, the indicator on the pack lights green.

3

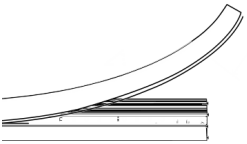


The run stays lit on the battery

Full charge takes 24 hours

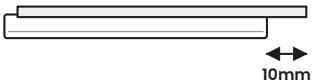
Emergency only: switch off the permanent live for a moment. The run must stay lit on the battery. Switch it back on to recharge.

4



Push the diffuser in from one end, a little at a time. **Never stretch it.** On a ring, start at a joint.

5



Leave 10mm spare at each end. Do not trim yet

Leave 10mm spare at each end. Do not trim yet.

6



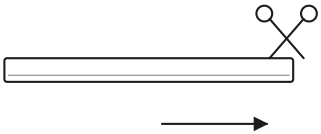
Give it time to settle before you trim

The more it was stretched, the longer it takes

The diffuser is **silicone**. If it got stretched while fitting, it will shrink back. **Give it time to settle before you trim**: the more it was stretched, the longer it needs. If in doubt, leave it overnight.

6 Test and fit the diffuser (continued)

7

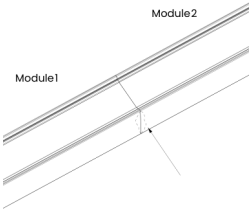


Trim, then push the ends under the end caps

Once it has settled, trim and push the ends under the end caps. If a gap opens later, ease it back along the run.

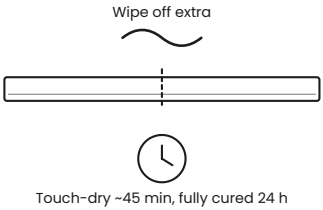
7 Seal the joints

1



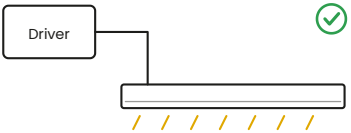
Once the fitting is fully installed and level, **seal both sides of every joint** with filling adhesive C38-00. Never skip this step.

2



Wipe off the extra. It is touch-dry in about 45 minutes. **Do not touch or clean the joint for 24 hours** while it fully cures.

3



Whole run lights evenly, no dim end
Fix any fault before sealing

Check from below with the lights on: no dark line or light leak at any joint.

8 Final check and troubleshooting

- [ ] Whole run lights evenly: no flicker, no dim end
- [ ] Dims smoothly through its full range (dimmable versions)
- [ ] Emergency: cut the permanent live, emergency section stays lit
- [ ] Emergency: indicator green, pack charged
- [ ] No light leaks or dark lines at joints
- [ ] Diffuser fully home at both ends, no gaps
- [ ] Level and straight; every fixing or wire taking load; flex slack
- [ ] This manual and driver details left with the client

Tick every box before you leave site.

8 Final check and troubleshooting (continued)

If something is wrong

What you see	Likely cause	What to do
Nothing lights	No mains at the driver, or + and - reversed	Check driver input, then output polarity
Part of the run dark	Tape damaged, or flex not fully into the input lead	Isolate; check the flex connection; inspect the tape
Far end dimmer than the feed end	Run over 5m fed from one end, or driver overloaded	Feed from both ends; check driver size
Flicker	Driver overloaded or wrong dimming type	Check driver size and /DD or /CC match
Bright spots through diffuser	Tape has come away	Re-press or re-stick the strip (section 4)
Gaps at diffuser ends	Diffuser stretched when fitted	Ease it back along the run
Dark line / light leak at a joint	Joint not sealed	Seal with C38-00 (section 7)
Emergency stays off	No permanent live, or battery not charged	Check the unswitched live; allow 24 hours to charge; retest

Still not right? Call 0161 877 1399 and **ask for Technical**, or email **technical@ricoman.com** with the order number and photos before taking anything apart.

Notes

Notes

Notes

Need help?

Call **0161 877 1399**

and ask for **Technical**

or email **technical@ricoman.com**

Have your order number and photos ready.

Watch the installation video

Scan the QR code with your phone camera, or go to

ricoman.com/flow-install

A quick 2-minute guide, plus the full step-by-step.

www.ricoman.com